

An Architectural Proposal for a Hybrid LLM-Augmented Shopping Recommender with Grounded Output

Prototype: **Zalem** · a working e-commerce store.

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Two tools, opposite weaknesses.

Classical recommenders

- + narrow a huge catalog accurately
- opaque; they never say **why**

LLM assistants

- + explain, summarize reviews, compare
- **hallucinate** prices, specs, even reviews

The benchmark and its documented failures.

32%

picks the right product

28%

prices are wrong

83%

favor its own brand

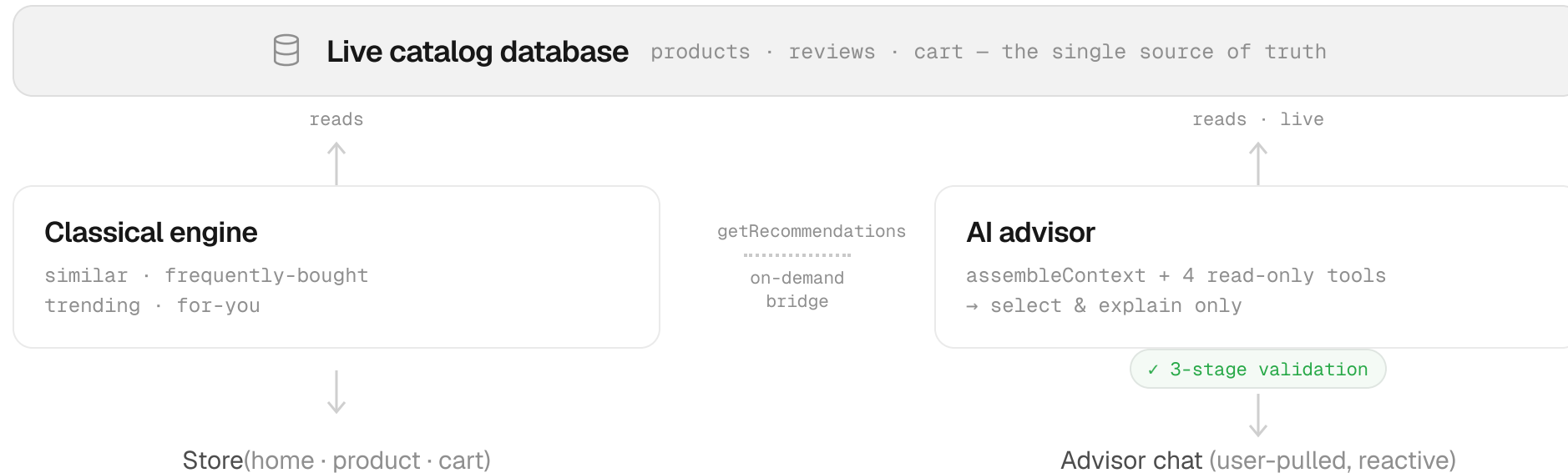
+

fabricated
review themes

Amazon Rufus — 300M+ users, ~\$12B sales · yet confidently wrong.

Grounded by construction: hydration over generation.

The model **selects and explains**; it never **sources** a fact. Every price, rating and spec is read from the live catalog, on every request.



The advisor won't make facts up.

-14%



AirPods Pro 2

★★★★★ (38)

1199 lei ~~1399 lei~~

 Add to cart

Shopping advisor

grounded

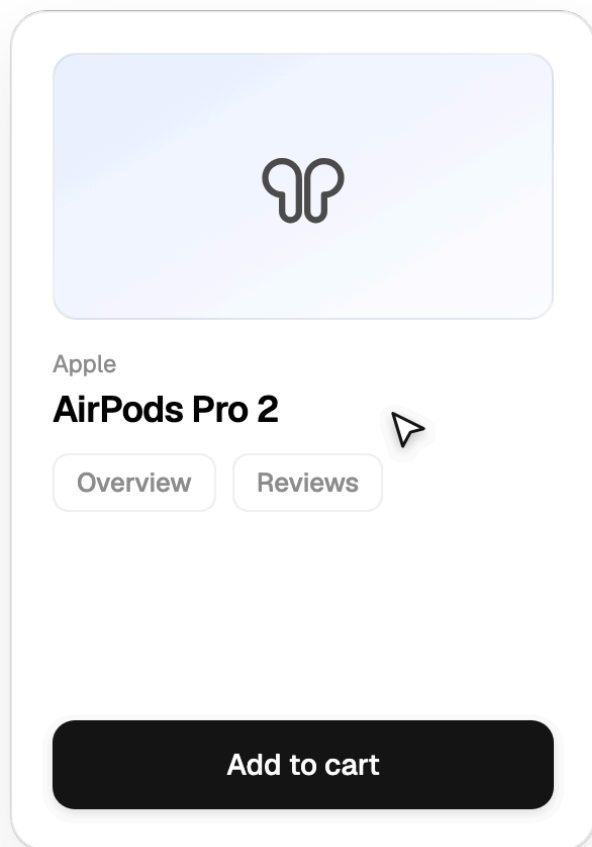
Are these good for the gym?

The AirPods Pro 2 are IPX4 sweat-resistant, and 11 reviewers call the fit secure for workouts. A few with smaller ears say they can work loose.

Ask about this product...



Grounded in the live session.



session state · rebuilt on every question

• Advisor context

assembled per question

CURRENT PRODUCT

AirPods Pro 2 · Apple
1199 lei · 4.6 ★ (38 reviews)

REVIEW THEMES

battery life (14) · comfortable fit (11)
divided — fit for small ears: 11 vs 5

CART

AirPods Pro 2 x1 · USB-C Cable x2

RECENTLY VIEWED

MacBook Air · iPad Air

⚡ Handed to the advisor with every question

Review summaries verified against the reviews.

- A nightly job extracts themes with real counts from every review.
- It surfaces genuine **conflicts** — who disagrees, and how many.
- Each theme is **embedding-verified** against the review set before it ships.

AirPods Pro 2 · what buyers think

38 reviews

LIKES

- Battery life 14
- Comfortable fit 11
- Noise cancellation 8

COMPLAINTS

- Touch controls 6

Divided on fit for small ears



11 secure

5 work loose

Does the grounding actually matter?

A controlled comparison: the **same model**, the same 34 questions — but with the catalog tools removed, answering from memory.

Metric	Grounded	Ungrounded
Specific price / rating claims	83	162
Claims that fail the catalog	48%	67%
Factuality	0.76	0.53

A `specific claim` = a concrete price or rating written in the answer (e.g. 1199 lei, 4.6/5) checked against the live database.

Trust is an integration problem, not a model-capability problem.

- Best out of 8 configs is the cheapest model: Gemini Flash-Lite · composite 0.807 · ~\$0.0003/query · p95 4.2s.
- **More reasoning was worse:** the big model at high reasoning was the worst (0.492) and slowest (12.1s p95).
- Grounding held everywhere: groundedness ≥ 0.89 across all 8 configurations.

Contributions and honest limits.

- **Contribution:** an integration architecture + an honest evaluation against documented production failures.
- Grounding is **structural and measured**.
- **Limits:** no user study · not tested against a standard product-search baseline · ~200-product seeded catalog.
- **Next:** a within-subjects user study on trust and purchase confidence · fine-tuning a smaller model

Demo.

Thank you.

Mihai-Dan Crisan · Questions.